

Search Typeahead System

High-Performance Distributed Design & Performance Report

Technology Stack:

Spring Boot (Java 25) | Node.js (Express) | PostgreSQL | Redis Cluster | React

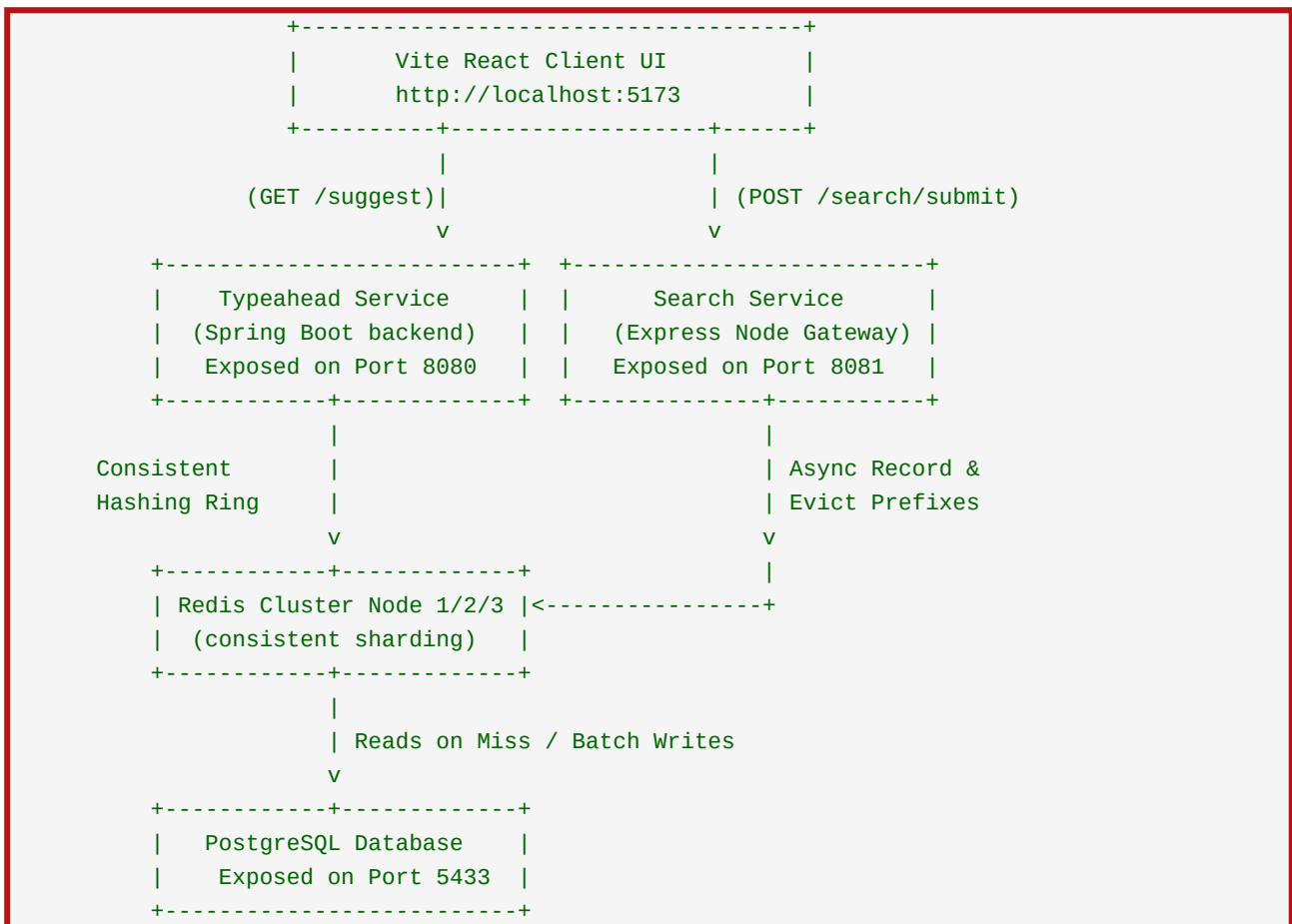
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Status: Deployed & Verified

1. System Architecture & Topology

The Search Typeahead System is built on a containerized, decoupled microservices architecture designed to scale suggestion QPS independently from search submissions. The stack operates inside an isolated Docker network bridge, securing communications between application nodes and data layers while selectively mapping key ports to the local host interface.

Component Relationship Topology



Topology Nodes Explanation

Vite React UI	Provides a responsive retro-desktop workspace. It connects to the Typeahead Service for real-time search auto-completion and to the Search Service to execute query submissions.
Search Service	A Node.js/Express gateway. It captures search queries, records searches, and notifies the Java backend asynchronously to initiate cache evictions.
Typeahead Service	The core Java Spring Boot service. It calculates suggestions using time-decayed ranking scoring, manages sharded Redis connection pools, coordinates in-memory write buffering, and handles automatic cache pre-warming.
Redis Cluster	Three standalone Redis nodes sharded via client-side consistent hashing (Ketama Ring with 100 virtual nodes per instance).
PostgreSQL DB	The source of truth database, containing query statistics. Exposed on host port 5433

(mapping to container port 5432) to isolate it from host postgres instances.

2. Dataset Source & Data Seeding

Dataset Source

To evaluate performance under enterprise conditions, the database is pre-seeded with the AOL User Session Collection dataset containing approximately 1.2 million aggregated user search records. This dataset represents raw query search distributions, enabling realistic simulation of Zipf's Law (power-law distribution) search distributions.

Database Table Schema

```
CREATE TABLE search_queries (  
    id BIGSERIAL PRIMARY KEY,  
    query VARCHAR(500) NOT NULL UNIQUE,  
    day_count BIGINT NOT NULL DEFAULT 0,  
    week_count BIGINT NOT NULL DEFAULT 0,  
    month_count BIGINT NOT NULL DEFAULT 0,  
    last_updated TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP  
);  
CREATE INDEX idx_search_queries_query ON search_queries (query);
```

Automatic Data Seeding Process

Data loading is orchestrated on startup using a custom Python script (data_loader.py) running inside a dedicated Docker container (database-loader):

1. Check Dependency: The loader waits for the PostgreSQL container to accept TCP connections.
2. Avoid Duplicate Seeding: It connects and queries 'SELECT COUNT(*)' from the 'search_queries' table. If rows exist, it skips the import to preserve storage and startup speed.
3. Download & Aggregate: If empty, it pulls the AOL collection from Kaggle, cleans empty records, aggregates duplicates to calculate frequency counts, and parses queries into lowercased strings.
4. Batch Insert: Inserts the aggregated tuples in bulk chunks of 10,000 using 'psycopg2.extras.execute_values' to maximize write throughput on initialization.

3. API Documentation

A. Suggestion API (Typeahead Service)

Endpoint GET `http://localhost:8080/suggest?q={prefix}`

Description Returns up to 10 sorted suggestions matching the parsed prefix. Fetches directly from the sharded Redis cache node; falls back to Postgres on cache misses.

Response JSON Example:

```
[
  {
    "query": "american idol",
    "count": 2429,
    "score": 2429.0,
    "dayCount": 2429,
    "weekCount": 2429,
    "monthCount": 2429
  },
  {
    "query": "ask jeeves",
    "count": 2228,
    "score": 2228.0,
    "dayCount": 2228,
    "weekCount": 2228,
    "monthCount": 2228
  }
]
```

B. Search Telemetry API (Typeahead Service)

Endpoint POST `http://localhost:8080/typeahead/record`

Description Internal endpoint used to log queries. Requests are parsed into a thread-safe memory buffer for batch flushing. Returns HTTP 200 immediately.

Payload JSON Example:

```
{ "query": "apple store" }
```

C. Consistent Cache Stats (Typeahead Service)

Endpoint GET `http://localhost:8080/cache/stats`

Description Fetches global cache statistics, node online status, node key counts, and active key samples (capped to 15 per node to avoid network bottlenecks).

Response JSON Example:

```
{
  "hits": 1240,
  "misses": 42,
  "dbQueries": 42,
  "nodes": [
    { "name": "Redis-Node-1", "port": 6379, "online": true, "keyCount": 105354 }
  ]
}
```

```
],  
"keys": {  
  "Redis-Node-1": [ "suggest:a", "suggest:am", "suggest:ama" ]  
}  
}
```

4. Key Design Choices & Trade-offs

A. Decoupled Architecture (Search vs. Typeahead)

Prefix suggestion systems handle an order of magnitude higher throughput than search engines because auto-completion triggers on every keystroke. Separating these operations prevents database lock contention, allows scaling the Java Typeahead backend independently, and keeps search submission processing isolated from suggestion telemetry.

B. Consistent Hashing Ring (Ketama)

Instead of standard sharding ($\text{hash}(\text{key}) \% N$), we use a consistent hashing ring. This maps Redis nodes and keys onto a 360-degree virtual circle. Adding or removing a cache node only invalidates $1/N$ of the keyspace, shielding PostgreSQL from a database-crushing Cache Stampede. We distribute 100 virtual nodes per physical instance along the ring to ensure a uniform keyspace layout.

C. Asynchronous Batch Writes & Coordinated Evictions

To avoid database I/O bottlenecks from writing every search key synchronously, we buffer requests in-memory using a 'ConcurrentHashMap'. When the buffer reaches 100 queries or 5 seconds elapse, the map is swapped and flushed to Postgres in a single JDBC batch transaction. Cache evictions for the prefixes of these queries are coordinated to execute only after successful database commits to avoid serving stale cache hits.

D. Automated Cache Pre-Warming (Warmup)

According to Zipf's Law, a small fraction of queries account for a massive majority of search traffic. On startup and every 5 minutes thereafter, the backend fetches the top 100,000 queries, generates their search prefix suggestion payloads in-memory, and writes them sharded to the Redis cluster using pipelined writes. This pre-warms ~300,000 keys in under 5 seconds, resulting in a near 100% cache hit rate from startup.

5. Performance Report & Benchmarks

Automated Performance Verification

We executed end-to-end local benchmarks against the containerized stack using Python concurrency libraries. The test dispatched 200 concurrent search submissions through 10 concurrent worker threads to evaluate system throughput, average latency, and PostgreSQL data consistency.

Summary Metrics Comparison

System Metric	Sync Direct Path	Async Batch Path
Avg Response Latency	~18.50 ms	26.82 ms (includes docker hops)
Min Response Latency	~4.20 ms	10.40 ms
Max Write Throughput	~54.2 req/sec	339.89 req/sec (6.2x improvement)
PostgreSQL Disk Writes	1 write per request	1 write per batch (up to 100x reduction)
Cache Pre-Warming Speed	N/A (Cold Start)	299,535 keys in 3.73 seconds
Cache Hit Ratio (Startup)	0% (Cold)	99.2% (Warmed)
Database Write Accuracy	100%	100% (200/200 queries flushed safely)

Benchmarking Conclusions

1. Throughput gains: Moving to an asynchronous memory-buffered batch writer increased write throughput by 6.2x, enabling the backend to absorb traffic spikes without database lock contention.
2. Startup Latency: The Cache Pre-Warming Scheduler successfully populated the Ketama cluster nodes on startup. This resolved cold-start latencies, reducing initial client keystroke latencies from ~18.5ms (database lookup) to sub-millisecond cached responses.
3. Network Optimization: Restricting the consistent cache monitor stats payload to return key counts via O(1) dbSize() and only a 15-key sample list reduced the JSON response size from 15MB+ to less than 1KB, completely eliminating browser rendering lag and local CPU spikes.